SAURABH RANJAN

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Ph.D. Cognitive Neuroscientist | Cognitive and Machine Psychology | Human-AI Alignment | NeuroAI

EDUCATION

Ph.D. Psychology — Behavioral & Cognitive Neuroscience University of Florida, USA | 2025

Dissertation: Reality Monitoring in Humans and Artificial Intelligence

M.Sc. Cognitive Science University of Allahabad, India | 2018

Thesis: Intentional Binding in Future-Directed Intentions

B.Sc. Physics Birla Institute of Technology, Mesra, India | 2016

TECHNICAL SKILLS

AI/ML & LLMs: Agentic workflows (LangChain/LangGraph), LLM evaluation & hallucination-detection pipelines, prompt engineering; PyTorch, Hugging Face, scikit-learn, Pydantic

AI Tooling & Infra: Claude Code (agent testing/tooling), Psych Scanner, MetaSignal, Read the Docs

Interpretability & NeuroAI: Representation probing, encoding models, MEG/EEG (MNE), fMRI (Nilearn/SPM), UK Biobank predictive modeling

Stats & Evaluation: Bayesian hierarchical modeling (PyMC, brms), signal detection theory, mixed-effects models (lme4), metacognitive modeling

Data Engineering & MLOps: Python, R, MATLAB, Bash; Pandas, NumPy, Git/GitHub, Docker, SLURM/HPC

Visualization: seaborn, matplotlib, plotly, ggplot2

RESEARCH EXPERIENCE

Researcher — AI for Biomedical Health Outcomes *2026 – Present*

University of Florida, Gainesville, FL

- Build agentic AI systems (planning, reflection, tool use, multi-agent coordination) for clinical decision-making and co-develop UF's Master's-level curriculum on Data Science and Agentic LLMs for AI for Biomedical Health Sciences (AIBHS) — spanning Foundations, Clinical Application, and Basic Science tracks and engineered with reproducible pipelines (PyTorch, LangChain/LangGraph, SLURM/HPC) — contributing to UF's #1 AI-readiness ranking among large public universities (AIREDEX, 2026); built and maintain the AIBHS Faculty Hub and AI Passport Impact Project sites (20 hands-on modules).
- Developed Psych Scanner-Primal, an optimized infrastructure for the Prime Intellect Environments Hub, for evaluating AI models under Reinforcement Learning with Verifiable Rewards (RLVR).

Research Assistant *2025 – 2026*

Dr. Andreas Keil's Lab, University of Florida, Gainesville, FL

- Continued development of Psych Scanner, advancing it from version 0.1.0 to version 0.3.0.

Graduate Researcher *2020 – 2025*

Dr. Brian Odegaard's PAC-Lab & UF Department of Psychology, Gainesville, FL

- Discovered that LLM source-attribution accuracy reverses under episodic memory delay, tested across 2 experiments and 6 transformer-based LLMs (Gemma3, Llama3.3, Llama4), a hallucination-relevant failure invisible to single-turn benchmarks.
- Found that corrective feedback produces two distinct metacognitive failure modes and that failure severity tracks active, not aggregate, parameter count, publishing the result as a preprint (arXiv:2607.23927).
- Designed and ran matched experiments with 100+ human participants and parallel LLM simulations, to generalize reality monitoring within human and LLM populations across task constraints.
- Applied neuroimaging decoding methods:
  - MEG temporal generalization to decode seen versus imagined experience
  - fMRI linearizing encoding models to predict fMRI response from image memorability in the NSD dataset
- Built psychological network models from 2,743 human participants and 6 LLMs (12B-272B parameters), showing human representational structure replicates across populations ($r = 0.31$-0.93) while LLMs fail to reproduce it at any scale, suggesting a world model can be characterized purely from the structural relationships between internally generated experiences in imagination and memory.
- Across these studies, LLMs showed distinct failures in knowing and accurately reporting their own knowledge, including at the largest scales tested (272B parameters) — a gap directly relevant to trusting and monitoring more capable future systems.
- Authored Psych Scanner, an open-source Python framework for running LLM cognitive experiments at scale.
- Mentored 7 undergraduate researchers and taught a full segment of Physiological Psychology.

Research Assistant *2019 – 2020*

Center of Behavioral and Cognitive Sciences, University of Allahabad, India

- Found stronger intentional binding, a measure of sense of agency, for predictive intermediate outcomes than non-predictive ones, replicated across three delays (300/500/700ms) and two contingency levels; published as an OSF preprint.
- This construct, monitoring one's own actions, now underlies how agentic AI systems track and self-correct across multi-step plans.

Research Assistant *2018*

Homi Bhabha Centre for Science Education, TIFR, Mumbai, India

- Contributed statistical analysis for Touchy-Feely Vectors (TFV), a gesture-based tool for teaching vectors, piloted across 266 grade-11 students (3 experimental, 3 control classrooms); the experimental group reasoned differently about vectors and showed higher engagement. Published in IEEE T4E 2019.
- Contributed statistical analysis for a 3-year field study of TFV (135 control vs. 131 experimental students), finding it improved model-based reasoning in stronger students and engagement in average ones. Published in the Journal of Computer Assisted Learning, 2021.

PUBLICATIONS

Working Papers

[1] **Ranjan, S.**, & Odegaard, B. Generalization of generation effect in reality monitoring.

Software

[2] [Preprint] **Ranjan, S.**, Makwana, M., Sokratous, K., & Odegaard, B. (2026). Metasignal: A python package for comprehensive metacognitive analysis and decision-making. arXiv:2607.29093. https://arxiv.org/abs/2607.29093

[1] **Ranjan, S.**, Sokratous, K., & Makwana, M. Psych Scanner: A Framework for Systematic Cognitive Evaluation of Large Language Models.

Blog

[1] **Ranjan, S.** (2026). "Why Time Feels the Way It Does." *Synthetic Selves* (Substack). https://syntheticselves.substack.com/p/why-time-feels-the-way-it-does

Theory and Empirical Peer-Reviewed & Preprints

[10] [Preprint] **Ranjan, S.,** Sokratous, K., & Odegaard, B. (2026). Reality Monitoring in Large Language Models: Self-Knowledge That Transforms with Conversation Memory. arXiv:2607.23927. https://arxiv.org/abs/2607.23927

[9] [Preprint] **Ranjan, S.**, & Odegaard, B. (2025). Psychological Imagination Networks Show Cross-Population Centrality and Clustering Alignment in Humans That Large Language Models Fail to Replicate. arXiv:2510.04391. https://arxiv.org/abs/2510.04391

[8] **Ranjan, S.**, & Odegaard, B. (2024). Heterarchy or hierarchy? Insights from a new model of visual imagination. *Physics of Life Reviews*, 49, 74–76.

[7] **Ranjan, S.**, & Odegaard, B. (2024). Reality monitoring and metacognitive judgments in a false-memory paradigm. *Neuroscience Research*, 201, 3–17.

[6] Maynes, R., Faulkner, R., Callahan, G., Mims, C. E., **Ranjan, S.**, Stalzer, J., & Odegaard, B. (2023). Metacognitive awareness in the sound-induced flash illusion. *Philosophical Transactions of the Royal Society B*, 378(1886), 20220347.

[5] Chiasson, P., Boylan, M. R., Elhamiasl, M., Pruitt, J. M., **Ranjan, S.**, Riels, K., . . . & Keil, A. (2023). Effects of neurofeedback training on performance in laboratory tasks: A systematic review. *International Journal of Psychophysiology*.

[4] Aggarwal, A., & **Ranjan, S.** (2022). How do undergraduate students reason about ethical and algorithmic decision-making? *Proc. 53rd ACM Technical Symposium on Computer Science Education*, 488–494.

[3] Karnam, D., Agrawal, H., Parte, P., **Ranjan, S.**, Borar, P., Kurup, P. P., . . . & Chandrasekharan, S. (2021). Touchy feely vectors: A compensatory design approach to support model-based reasoning. *Journal of Computer Assisted Learning*, 37(2), 446–474.

[2] Karnam, D., Agrawal, H., Parte, P., **Ranjan, S.**, Sule, A., & Chandrasekharan, S. (2019). Touchy feely affordances of digital technology for embodied interactions can enhance 'epistemic access'. *2019 IEEE Tenth International Conference on Technology for Education (T4E)*, 114–121.

[1] [Preprint] **Ranjan, S.**, & Srinivasan, N. (2019). Sense of agency for future-directed intentions. doi.org/10.31234/osf.io/qa93k.

INVITED TALKS

[2] Ranjan, S. (May, 2026). *The Spark of Artificial Neuroscience: Psychologically-Grounded Evaluation of Large Language Models*. Virtual invited talk, Autonomous Empirical Research Group + Laboratory for Automated Scientific Discovery of Mind and Brain (PI: Dr. Sebastian Musslick), Osnabrück University.

[1] Ranjan, S. (March, 2026). *Controlling Imagery Generation and its Awareness.* Virtual invited talk, Robert Reinhart Lab, Boston University.

SELECTED CONFERENCE POSTERS & TALKS

[8] **Ranjan, S.**, & Odegaard, B. (2025). Psychological Imagination Networks in Humans and LLMs. Frontiers in NeuroAI, Kempner Institute Symposium, Harvard University.

[7] **Ranjan, S.**, & Odegaard, B. (2025). Visual Imagination Networks in Humans and LLMs. Vision Sciences Society Annual Meeting, St. Pete, FL.

[6] **Ranjan, S.**, & Odegaard, B. (2024). The Fragility of Reality Monitoring under Extraneous Factors. Psychonomic Society 65th Annual Meeting, NYC.

[5] **Ranjan, S.**, & Odegaard, B. (2024). Reality Monitoring, Fast and Slow [Poster + Flash Talk]. Association for Psychological Science, San Francisco. *Awarded Scott O. Lilienfeld APS Travel Award.*

[4] Maw, M., Zhuang, L., Baltes, J., **Ranjan, S.**, & Odegaard, B. (2024). Task demands and sensory externalization in reality monitoring. PGSO Undergraduate Research Forum, UF. *Mentee won 3rd Prize.*

[3] Dundigalla, S., Johnson, D., Roh, A., **Ranjan, S.**, & Odegaard, B. (2024). Cognitive strategies in reality monitoring. PGSO Undergraduate Research Forum, UF.

[2] **Ranjan, S.**, Baltes, J., Roh, A., & Odegaard, B. (2023). Confidence in reality monitoring judgments. *Journal of Vision*, 23(9), 4844.

[1] **Ranjan, S.**, & Srinivasan, N. (2018). Intentional binding and future-directed intentions [Talk]. Annual Conference of Cognitive Science, IIT-Guwahati, India.

AWARDS & FELLOWSHIPS

- 2025 — College of Liberal Arts and Sciences Travel Award, University of Florida
- 2025 — Threadgill Dissertation Fellowship, University of Florida
- 2024 — Scott O. Lilienfeld APS Travel Award, Association for Psychological Science
- 2024–2025 — UF Department of Psychology Travel Awards (×4)
- 2020–2025 — Graduate Student Fellowship, UF Department of Psychology
- 2016–2018 — Graduate Merit Scholarship, Centre of Behavioral & Cognitive Sciences, University of Allahabad